4.1. PRICING KERNELS

To capture the composite bad times over multiple factors, the new asset pricing approach uses the notion of a *pricing kernel*. This is also called a *stochastic discount factor* (SDF). We denote the SDF as m. The SDF is an index of bad times, and the bad times are indexed by many different factors and different states of nature. Since all the associated recent asset pricing theory uses this concept and terminology, it is worth spending a little time to see how this SDF approach is related to the traditional CAPM approach. There is some nice intuition that comes about from using the SDF, too. (For the less technically inclined, you are welcome to skip the next two subsections and start again at section 4.3.)

By capturing all bad times by a single variable m, we have an extremely powerful notation to capture multiple definitions of bad times with multiple variables. The CAPM is actually a special case where m is linear in the market return:[5]

$$m = a + b \times r_m, \tag{6.3}$$

for some constants a and b. (A pricing kernel that is linear in the market gives rise to a SML that with asset betas with respect to the market in equation (6.2).) With our "m" notation, we can specify multiple factors very easily by having the SDF depend on a vector of factors, $F = (f_1, f_2, \ldots, f_K)$:

$$m = a + b_1 f_1 + b_2 f_2 + \ldots + b_K f_K, \tag{6.4}$$

where each of the K factors themselves define different bad times.

Another advantage of using the pricing kernel m is that while the CAPM restricts m to be *linear*, the world is *nonlinear*. We want to build models that capture this nonlinearity.[6] Researchers have developed some complicated forms for m, and some of the workhorse models that we discuss in chapters 8 and 9 describing equities and fixed income are nonlinear.

[5] The constants a and b can be derived using equation (6.3) as

$$a = \frac{1}{1 + r_f} + \frac{\mu_m(\mu_m - r_f)}{(1 + r_f)\sigma_m^2},$$

$$b = -\frac{\mu_m - r_f}{(1 + r_f)\sigma_m^2},$$

where $\mu_m = E(r_m)$ and $\sigma_m^2 = \mathrm{var}(r_m)$ are the mean and variance of the market returns, respectively. Note that the coefficient b multiplying m is negative: low values of the SDF correspond to bad times, which in the CAPM are given by low returns of the market.

[6] Related to this is that the requirement for the SDF is very weak: it requires only no arbitrage, as shown by Harrison and Kreps (1979). The CAPM, and other specific forms of m, on the other hand, require many additional onerous and often counterfactual assumptions.